

Name		Index No.		Civics Group	
------	--	-----------	--	--------------	--



DUNMAN HIGH SCHOOL
SENIOR HIGH
 2009 PRELIMINARY EXAMINATION
 General Certificate of Education Advanced Level
 YEAR SIX H2 BIOLOGY (9747)

BIOLOGY

9747 / 01

PAPER 1 Multiple Choice

25 September 09 (Friday)

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in the **soft pencil** on separate Answer Sheet [**optical answer sheet (OTAS)**] provided.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

Do not open this booklet until you are told to do so.

This question paper consists of **18** printed pages and **2** blank pages (including this cover page).

Section A: Multiple Choice Questions (40 marks)

Answer all questions.

- 1** Which molecules, found in cell surface membranes, contribute to cell recognition?
- A** cholesterol and glycolipids
- B** cholesterol and phospholipids
- C** glycolipids and glycoproteins
- D** phospholipids and glycoproteins
- 2** Equal quantities of a 5% starch solution were added to equal quantities of different solutions.

After 5 minutes at room temperature, the solutions were tested with iodine in potassium iodide solution and the depth of colour was recorded using a colorimeter (with an arbitrary scale of 1 to 5, 5 being the darkest blue).

The results are given in the table.

	solution	colorimeter reading
1	saliva and dilute HCl	4
2	saliva and dilute NaHCO ₃	1
3	dilute HCl only	4
4	saliva at 100°C	5
5	saliva at 20°C	3

This experiment suggests that saliva contains a biologically active substance which _____.

- A** hydrolyses starch in acid conditions
- B** is denatured by hydrochloric acid
- C** is promoted in its action by high temperatures
- D** may be activated by hydrogencarbonate ions

3 Which of the following food samples requires the largest amount of oxygen to oxidize it completely?

- A** 1 g of amino acid
- B** 1 g of fat
- C** 1 g of monosaccharide
- D** 1 g of polysaccharide

4 Which is the longest phase in the cell cycle of human liver cells?

- A** prophase
- B** anaphase
- C** cytoplasmic cleavage
- D** interphase

5 Down's syndrome can exist in two forms.

Classic Down's syndrome affects all cells, while the mosaic form arises from the presence of two or more cell types in the body with differences in the chromosome number and structure.

How does the mosaic condition arise?

- A** Non-disjunction of chromosomes in mitosis in early foetal development
- B** Non-disjunction of chromosomes in mitosis during formation of ova
- C** Non-disjunction of chromosomes in mitosis during formation of sperm
- D** Translocation of chromosomes at maturation of ovum or the sperm

- 6 An experiment is carried out to show that DNA replication is semi-conservative. *Escherichia coli* cells whose DNA contains only 'heavy' nitrogen (^{15}N) are grown in a culture medium containing the normal isotope of nitrogen (^{14}N) and allowed to divide three times.

What will be the percentage of DNA molecules containing both forms of nitrogen after the third division?

- A 12.5%
- B 25%
- C 50%
- D 100%

- 7 Pyrimidine bases contain 4 carbon atoms and purine bases contain 5.

How many carbon atoms does adenosine triphosphate contain?

- A 8
- B 9
- C 10
- D 11

- 8 Transfer RNA _____.

- A forms hydrogen bonds between its codon and the anticodon of an mRNA in the A site of a ribosome
- B binds to its specific amino acid in the active site of an aminoacyl-tRNA synthetase
- C uses GTP as the energy source to bind its amino acid
- D is produced in the nucleolus

9 Which of the following correctly describes the characteristics of a virus?

✓ represents presence of characteristic

✗ represents absence of characteristic

	obligate parasitism	involvement of host enzymes	involvement of host cell ribosomes	entry of nucleic acid into a cell
A	✗	✗	✓	✗
B	✓	✗	✓	✓
C	✓	✓	✗	✗
D	✓	✓	✓	✓

10 Scientists have discovered how to put together a bacteriophage with the protein coat of phage T2 and the DNA of phage T4. If this composite phage were allowed to infect a bacterium, the phages produced in the host cell would have _____.

- A the protein of T2 and the DNA of T4
- B the protein and DNA of T4
- C the protein of T4 and the DNA of T2
- D the protein and DNA of T2

11 An Hfr bacterium is one that has _____.

- A a special recognition site that will take up closely related DNA from its environment
- B several insertion sequences scattered throughout its chromosome
- C several copies of a single transposon repeated randomly throughout its chromosome
- D a plasmid that has become integrated into its chromosome

- 12 The table shows a comparison between the genomes of a prokaryote and a eukaryote.

	prokaryote	eukaryote
genome length (base pairs)	4,640,000	12,068,000
number of proteins	4,300	6,200
proteins with roles in metabolism	650	650
proteins with roles in energy release	240	175
proteins with roles in protein synthesis	410	750

Which feature of eukaryotes accounts for the differences in the number of proteins with roles in protein synthesis?

- A The DNA of a eukaryote has histones.
 - B The DNA of a eukaryote has introns.
 - C The DNA of a eukaryote has more base pairs.
 - D The DNA of a eukaryote is separated from the ribosomes by membranes.
- 13 Which best describes the correct order (from earliest to latest) in the control of gene expression in protein synthesis?
- A Polyribosome, promoter, spliceosome, primary transcript
 - B Spliceosome, promoter, primary transcript, polyribosome
 - C Promoter, primary transcript, spliceosome, polyribosome
 - D Primary transcript, spliceosome, promoter, polyribosome
- 14 How is the coordinated transcription of genes involved in the same pathway regulated?
- A The genes are transcribed in one transcription unit, although each gene has its own promoter.
 - B The genes are located in the same region of the chromosome, and enzymes deacetylate the entire region so that transcription may begin.
 - C The genes all respond to the same general transcription factors, though they may respond to different specific transcription factors.
 - D The genes have the same combination of control elements in the enhancer that bind with the particular activators present in the cell.

- 15 In a dihybrid cross of heterozygotes, what proportion of the offspring will be phenotypically dominant for both traits?

A 1/16
 B 3/16
 C 1/4
 D 9/16

- 16 The base height of the dingdong plant is 10 cm. Four genes contribute to the height of the plant, and each dominant allele contributes 3 cm to height.

If you cross a 10 cm plant (quadruply homozygous recessive) with a 34 cm plant, how many phenotypic classes will there be in the F_2 ?

A 4
 B 5
 C 8
 D 9

- 17 Four bacteriophage genes were mapped by recombination. The recombination frequencies between the four genes are shown in the table below.

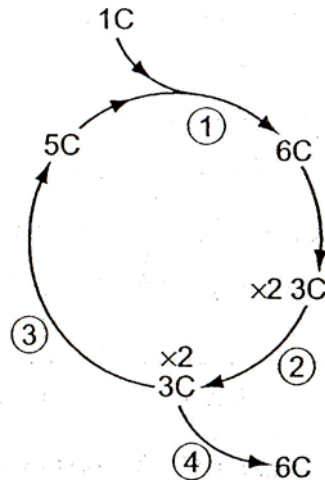
	A	B	C	D
A	-	0.18	0.10	0.20
B		-	0.15	0.37
C			-	0.24
D				-

What is the order of the genes along the chromosome?

A A, B, C, D
 B C, B, D, A
 C B, A, D, C
 D D, A, C, B

- 18** During aerobic respiration, most ATP production occurs during the _____.
- A** conversion of pyruvate to lactate
 - B** conversion of triose phosphate to pyruvate
 - C** conversion of hexose phosphate to triose phosphate
 - D** oxidative decarboxylation of pyruvate to CO_2 and H_2O
- 19** What happens to most of the reduced nicotinamide adenine dinucleotide (NADH_2) molecules in cell metabolism?
- A** oxidation in the Calvin cycle
 - B** oxidation in mitochondrial ATP formation
 - C** combination with succinic acid as part of the Krebs cycle
 - D** acting as an oxidising agent for FADH_2

- 20 The diagram shows some phases of the Calvin cycle and the numbers of carbon atoms in the compounds involved.



Which reactions occur at phases 1 to 4?

A

Phase 1	RuBP + CO ₂
Phase 2	ATP → ADP + P _i NADP → reduced NADP
Phase 3	ATP → ADP + RuBP
Phase 4	triose → hexose

B

Phase 1	triose → hexose
Phase 2	ADP + P _i → ATP NADP → reduced NADP
Phase 3	ADP + RuBP → ATP
Phase 4	glycerate 3-phosphate → triose

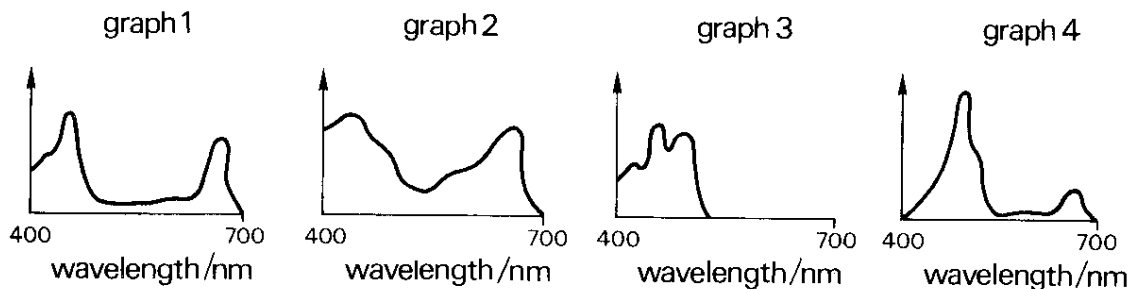
C

Phase 1	triose → hexose
Phase 2	ADP + P _i → ATP reduced NADP → NADP
Phase 3	ADP + RuBP → ATP
Phase 4	glycerate 3-phosphate → triose

D

Phase 1	RuBP + CO ₂
Phase 2	ATP → ADP + P _i reduced NADP → NADP
Phase 3	ATP → ADP + RuBP
Phase 4	triose → hexose

- 21 In a classic experiment on photosynthesis, R. L. Hill demonstrated that an illuminated *in vitro* suspension of isolated chloroplasts could produce oxygen in the presence of a hydrogen acceptor such as methylene blue. In this case methylene is reduced. Which one of the following compounds replaces methylene blue in the intact photosynthesising plant?
- A adenosine triphosphate (ATP)
 B carbon dioxide
 C nicotinamide adenine dinucleotide phosphate (NADP)
 D ribulose biphosphate (RuBP)
- 22 Three of the graphs below show the absorption spectra of photosynthetic pigments. One graph shows the action spectrum of photosynthesis for a plant containing the pigments.

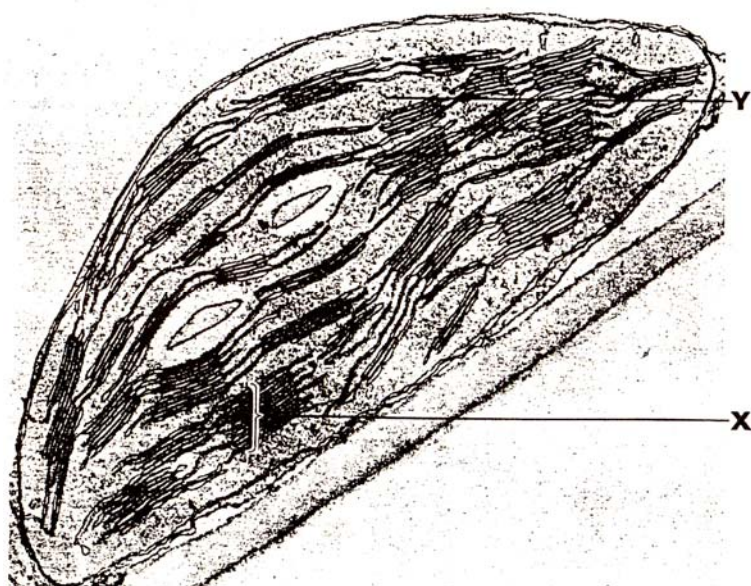


All the x axes show wavelength. Three of the y axes show light absorption. One y axis shows the rate of photosynthesis.

Which of the following identifies the four graphs?

Absorption spectra				
	Chlorophyll a	Chlorophyll b	Carotenoids	Action spectrum
A	1	4	3	2
B	2	1	3	4
C	3	2	4	1
D	4	2	1	3

23



What are the structures labelled X and Y and what are their main functions?

	Structure X	Function of X	Structure Y	Function of Y
A	granum	photolysis of water	stroma	fixation of CO ₂
B	granum	fixation of CO ₂	lamella	photolysis of water
C	lamella	fixation of CO ₂	granum	photolysis of water
D	stroma	photolysis of water	granum	fixation of CO ₂

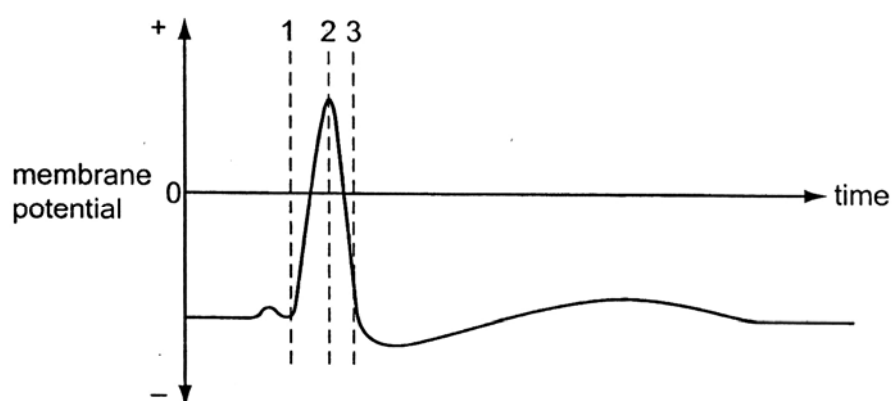
24 All of the following are examples of a negative feedback system **except** _____.

- A** decreases in blood glucose brought about by increases in insulin secretion by the pancreas
- B** increases in blood glucose brought about by increases in glucagon secretion by the pancreas
- C** increases in milk production by mammary glands brought about by increasingly more nursing by a baby
- D** increase in sweat production brought about by the hypothalamus in response to increasing body temperatures

25 Which axon would transmit an action potential most rapidly?

- A 1 mm diameter neurone lacking myelin.
- B 1 mm diameter neurone with myelin.
- C 2 mm diameter neurone lacking myelin.
- D 2 mm diameter neurone with myelin.

26 The diagram shows the changes in potential of the axoplasm relative to the outer surface of the axon at a point on an axon during the passage of a nerve impulse.

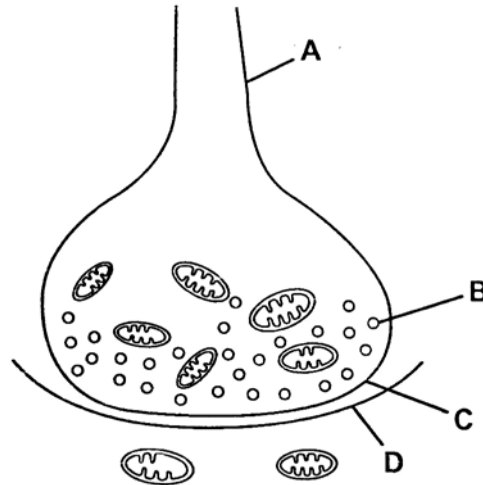


Which events are occurring between times 1, 2 and 3 to produce the action potential illustrated?

	between 1 and 2	between 2 and 3
A	inflow of potassium ions	outflow of sodium ions
B	inflow of sodium ions	outflow of potassium ions
C	outflow of potassium ions	inflow of sodium ions
D	outflow of sodium ions	inflow of potassium ions

- 27 The diagram shows the structure of a synapse.

Which membrane can produce either a depolarisation or a hyperpolarisation in response to specific chemical stimuli?



- 28 Dinitrophenol is a metabolic poison that prevents the production of ATP.

What will happen if a resting axon is treated with such a poison?

- A The charge inside the axon membrane becomes more negative.
 - B The charge inside the axon membrane becomes less positive.
 - C The charge outside the axon membrane becomes less positive.
 - D The charge outside the axon membrane does not change.
- 29 Which of the following is **not** true about hormones?
- A Hormones are chemical messengers that travel through the circulatory system.
 - B Hormonal secretion is independent of influence from the nervous system.
 - C Hormonal action is often regulated through feedback loops.
 - D Hormones often regulate homeostasis through antagonistic functions.

30 How does insulin act on its target cell?

- A** It activates enzyme conversion of glycogen to glucose.
- B** It alters specific receptor sites on the cell surface membrane.
- C** It enters the cell and stimulates transcription of DNA.
- D** It stimulates the intracellular hydrolysis of lipids.

31 Signal pathways involve components called secondary messengers, characteristics of these substances are as follows:

- i. small
- ii. non-protein
- iii. water soluble molecules / ions
- iv. enzymatic in nature
- v. membrane bound

Which of the following correctly describes secondary messengers?

- A** i, ii and iii
- B** i, ii and v
- C** i, iii and v
- D** i, ii, iii and v

32 How do receptor tyrosine kinases transduce a signal?

- A** They transport the signal molecule into the cell, where it binds to and activates a transcription factor. The transcription factor then alters gene expression.
- B** Their activated tyrosine kinases convert ATP to cAMP; cAMP acts as a second messenger to activate other protein kinases.
- C** Signal binding causes a conformational change that activates membrane-bound tyrosine kinase relay proteins that phosphorylate serine and threonine amino acids.
- D** They form a dimer; they phosphorylate each other's tyrosines; specific proteins bind to and are activated by these phosphorylated tyrosines.

- 33** Which one of the following statements describes a feature of a natural system of classification which an artificial system does **not** have?
- A** A natural system of classification is based on evolutionary relationships between organisms.
 - B** A natural system of classification is based on cytological characters.
 - C** A natural system of classification is based on single characteristics rather than several similar features.
 - D** A natural system of classification is based on phenotypic characters.
- 34** The average size of some species of birds found in cold climates is greater than members of the same species in warmer climates. A likely explanation of this observation is that _____.
- A** higher temperatures inhibit growth of the birds
 - B** the birds eat more in the cold climates
 - C** large animals have higher metabolic rates
 - D** in warm climates, birds with high metabolic activity are eliminated

- 35** A eukaryotic gene has "sticky ends" produced by the restriction endonuclease EcoRI. The gene is added to a mixture containing EcoRI and a bacterial plasmid that carries two genes, which make it resistant to ampicillin and tetracycline. The plasmid has one recognition site for EcoRI located in the tetracycline resistance gene.

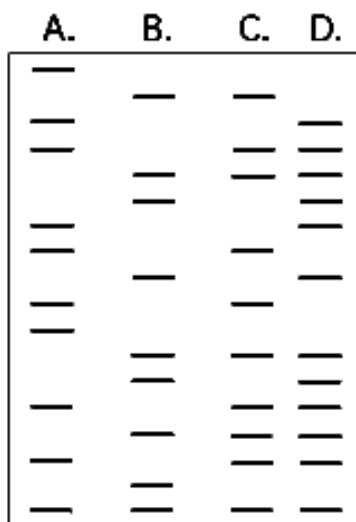
This mixture is incubated for several hours and then added to bacteria growing in nutrient broth. The bacteria are allowed to grow overnight and are streaked on a plate using a technique that produces isolated colonies that are clones of the original. Samples of these colonies are then grown in four different media:

- (i) nutrient broth plus ampicillin;
- (ii) nutrient broth plus tetracycline;
- (iii) nutrient broth plus ampicillin and tetracycline;
- (iv) and nutrient broth containing no antibiotics.

The bacteria that contained the plasmid, but not the eukaryotic gene, would grow _____.

- A** in the nutrient broth plus ampicillin, but not in the broth containing tetracycline
- B** in the broth containing tetracycline, but not in the broth containing ampicillin
- C** only in the broth containing both antibiotics
- D** in all four types of broth

36 The DNA profiles below represent four different individuals.



Which of the following statements is consistent with the results?

- A** B is the child of A and C.
- B** C is the child of A and B.
- C** D is the child of B and C.
- D** A is the child of B and C.

37 From the list below, which of the following is the most logical sequence of steps for splicing foreign DNA into a plasmid and inserting the plasmid into a bacterium?

- I. Transform bacteria with recombinant DNA molecule
- II. Cut the plasmid DNA using restriction enzymes
- III. Extract plasmid DNA from bacterial cells
- IV. Hydrogen-bond the plasmid DNA to nonplasmid DNA fragments
- V. Use ligase to seal plasmid DNA to nonplasmid DNA

- A** I, II, IV, III, V
- B** III, IV, V, I, II
- C** III, II, IV, V, I
- D** II, III, V, IV, I

- 38 Which of the following statements is **false** about haematopoietic stem cells?
- A Haematopoietic stem cells are able to reproduce continually.
 - B Haematopoietic stem cells can be found in the bone marrow.
 - C Haematopoietic stem cells can differentiate into all blood cell types.
 - D Haematopoietic stem cells can differentiate into all specialised cells.
- 39 Somatic hybrids are produced through _____.
- A asexual fusion
 - B protoplasmic fusion
 - C vegetative propagation
 - D grafting
- 40 What is a “pharm” animal?
- A a transgenic animal that produces large quantities of a pharmaceutical product
 - B an animal used by the pharmaceutical industry to test new medical treatments
 - C a cloned animal that was produced from an adult cell nucleus inserted into an ovum
 - D a genetically engineered animal that produces more meat or milk

END OF PAPER

Name		Index No.		Civics Group	
------	--	-----------	--	--------------	--



DUNMAN HIGH SCHOOL SENIOR HIGH

2009 PRELIMINARY EXAMINATION

General Certificate of Education Advanced Level

YEAR SIX H2 BIOLOGY (9747)

BIOLOGY

9747 / 02

PAPER 2 Core Paper

16 September 09 (Wednesday)

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Do not open this booklet until you are told to do so.

Section A:

Consists of 8 Structured Questions

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section B:

Consists of 2 Essay Questions

Answer any **one** question.

Write your answers on the separate **writing papers** provided.

At the end of the examination, fasten all your work securely together.

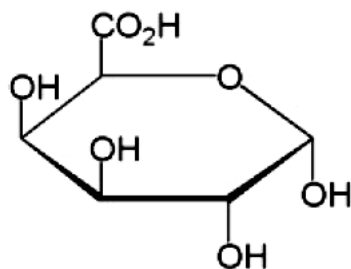
The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE		
Section A		/40
Section B	1	/11
	2	/16
	3	/10
	4	/10
	5	/10
	6	/10
	7	/ 8
	8	/ 5
Section C	9 / 10 *circle accordingly	/20
Total		

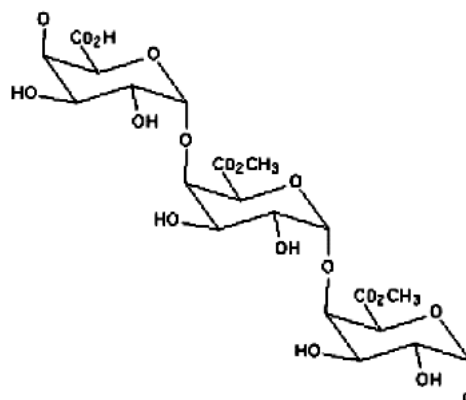
This question paper consists of **21** printed pages and **1** blank page (including this cover page).

Section B: Structured Questions (80 marks)Answer **all** questions in this section.**Question 1**

- (a) The structure of pectin, one of the polysaccharide components present in plant cell walls, is shown in **Figure 1.1** below. The monomer of pectin is D-galacturonic acid.



D-galacturonic acid



pectin

Figure 1.1

- (i) State **two** structural differences between pectin and cellulose. [2]

An important aspect of the ripening process of the tomato fruit is softening. The level of the enzyme polygalacturonase (PG) increases rapidly in the cell wall during ripening.

(ii) Describe briefly how PG leads to tissue softening during ripening. [1]

In countries which do not have well-developed refrigeration systems, spoilage due to fruit ripening contributes significantly to post-harvest losses.

(iii) Suggest **two** ways to delay the ripening process without relying on refrigeration. [2]

- (b) The diagram below shows the structure of haemoglobin. It consists of two α chain subunits and two β chain subunits. Each subunit is a conjugated protein associated with a haem group which is the site for oxygen transport. Each haem group in turn contains an atom of Fe^{2+} at its centre.

The haemoglobin molecule is a good transport protein for oxygen as the binding of oxygen to one haem group facilitates the binding of oxygen to the other haem groups on the same haemoglobin molecule. Similarly, when one haem group unloads its oxygen, the other three follow the lead and unload their oxygen as well. This is known as *cooperative oxygen binding and release*.

The binding of hydrogen carbonate (derived from carbon dioxide that dissolves in the blood) and carbon monoxide reduces the affinity of haemoglobin for oxygen. The hydrogen carbonate binding site is shown in the diagram; the carbon monoxide binding site corresponds to that of the oxygen binding site on the haem group.

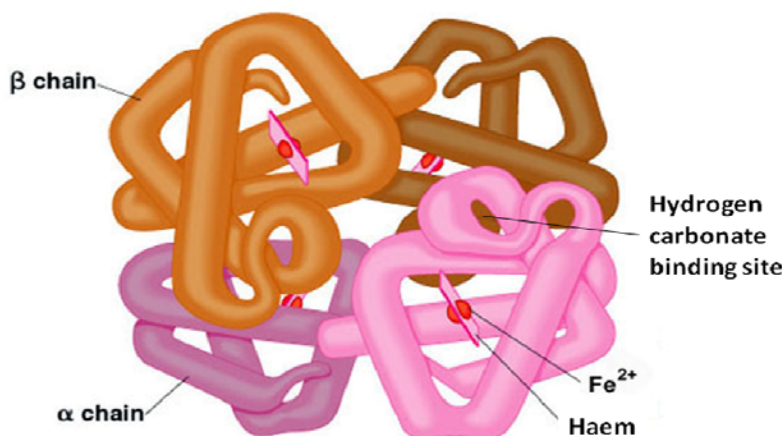


Figure 1.2

- (i) With reference to Figure 1.2, what are the evidences that the protein has tertiary and quaternary structure? [2]

- (ii) Suggest, in terms of molecular structure, why *cooperative oxygen binding and release* occurs. [2]

- (iii) Contrast the mechanisms by which hydrogen carbonate and carbon monoxide cause a reduction in oxygen affinity. [2]

Total: [11]

Question 2

(a) Explain how the following cellular organelles are adapted for genetic expression:

(i) nucleus; and [4]

(ii) ribosome [4]

(b) Describe the structure of a DNA nucleotide. [3]

(c) Describe how pre-mRNA is processed into mRNA. [3]

(d) Describe one way that a control element may influence transcription. [2]

Total: [16]

Question 3

(a) State three ways in which viruses differ structurally from prokaryotes. [3]

(b) Outline the events that occur during the replication of an Influenza A virus. [5]

(c) Suggest how the envelope which surrounds the influenza virus permits entry to host cells. [2]

Total: [10]

Question 4

Mutations also occur in human genes. A rare form of rickets that is not successfully treated with vitamin D therapy is caused by a mutant allele on the X chromosome. Rickets is a childhood disorder involving softening and weakening of the bones. It can be caused by a lack of vitamin D, calcium or phosphate.

Figure 4.1 shows a pedigree chart for a family that has a history of this condition.

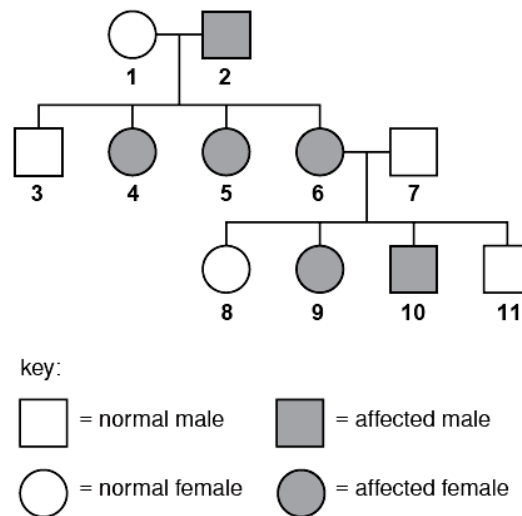


Figure 4.1

- (a) Using the symbols X^R = mutant allele on X chromosome
 X^r = normal allele on X chromosome,

state the genotype of the following individuals. [3]

1

3

9

- (b) The gene in which this mutation occurs, codes for a protein carrier found in the proximal convoluted tubule of the kidney. This protein is involved in phosphate transport across membrane.

Suggest why individuals with this mutant allele show symptoms of rickets. [2]

Coat colour in rabbits is determined by a single gene which has four separate alleles.

The gene is **not** sex linked.

- The allele for agouti colour, C^A , is dominant to all the other alleles.
- The allele for albino, C^a , is recessive to all the other alleles.
- The allele for chinchilla, C^{ch} , is dominant to the Himalayan allele, C^h .

- (c) (i) State all the possible genotypes for the following phenotypes. [2]

chinchilla

agouti

A young girl owns a pet female Himalayan rabbit. She wants to know whether it is homozygous (pure breeding) for this trait. Her friend owns a male albino rabbit, and says that if they cross the two rabbits and they find any albino offspring she can be sure that the female is not pure breeding.

- (ii) Name the type of cross that they will carry out. [1]

(iii) The friend's statement is valid. Explain why.

You may use genetic diagrams in your answer. [2]

Total: [10]

Question 5

Figure 5.1 is a summary of the light independent stage reactions of photosynthesis (Calvin cycle).

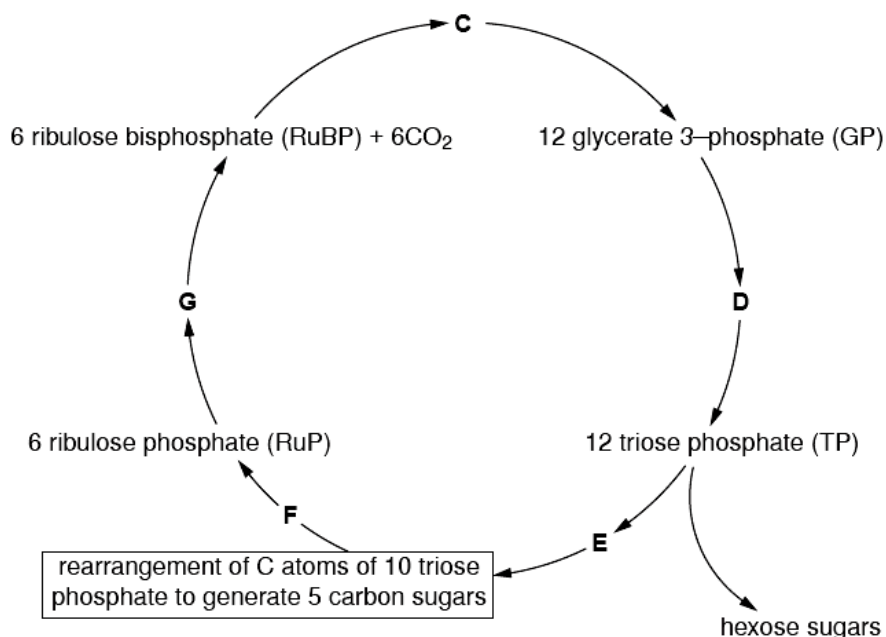


Figure 5.1

- (a) (i) Name the two products of the **light dependent** stage of photosynthesis that are required in the Calvin cycle. [1]

1

2

- (ii) Using Figure 5.1, state the letters of the two steps where chemical named in (a)(i) are required. [1]

1

2

- (iii) State two possible fates of the hexose sugars shown in Figure 5.1. [1]

1

2

- (b)** Describe and explain what will happen to the concentrations of ribulose biphosphate (RuBP) and glycerate-3-phosphate (GP) if carbon dioxide concentration falls. [2]

The rate of photosynthesis is determined mainly by environmental limiting factors. These are light intensity, the availability of carbon dioxide and temperature. Water supply has indirect effects by influencing the availability of carbon dioxide.

- (c)** Define the term 'limiting factor'. [1]

- (d)** Explain how water shortage could have an indirect effect on photosynthesis by influencing the availability of carbon dioxide. [2]

Some plants, such as wood sorrel, *Oxalis acetosella*, nearly always grow in shade where light intensity is commonly a limiting factor for photosynthesis. They are known as shade plants. Plants that live in open habitats, for example the daisy, *Bellis perennis*, are called sun plants.

Figure 5.2 shows the net rate of photosynthesis of sun and shade plants in response to increasing light intensity. The net rate of photosynthesis is defined as:

Mass of CO₂ fixed in photosynthesis minus mass of CO₂ produced in respiration, per unit time.

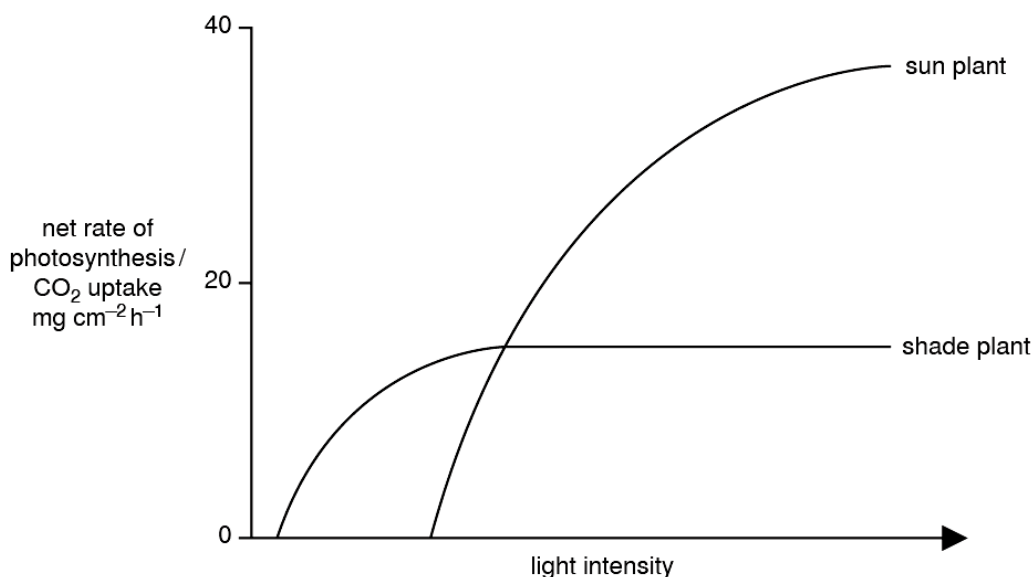


Figure 5.2

- (e) Using Figure 5.2, describe the responses of sun plant and shade plant to increasing light intensity. [2]

Total: [10]

Question 6

The first stage in respiration involves the conversion of one molecule of glucose into two molecules of a 3C compound that can enter mitochondria when oxygen is present.

(a) State

(i) the name of the compound that enter the Krebs cycle; [1]

(ii) the energy transfer that takes place in respiration; [1]

(iii) the function of this energy transfer. [2]

(b) Reduced NAD is another product of the first stage of respiration.

Describe the role and fate of reduced NAD in respiring yeast cells in the absence of oxygen.[3]

- (c) In mammals, anaerobic respiration in muscle tissue results in the production of lactate.

Describe how the production of lactate in muscle tissue differs from anaerobic respiration in yeast. [3]

Total: [10]

Question 7

In an experiment to investigate the effects of insulin on glucose uptake by muscle cells, the concentration of free glucose inside muscle cells was measured with respect to the extracellular glucose concentration, in the presence of insulin (labelled as insulin) and in the absence of insulin (labelled as control). The results are illustrated in Figure 7.1.

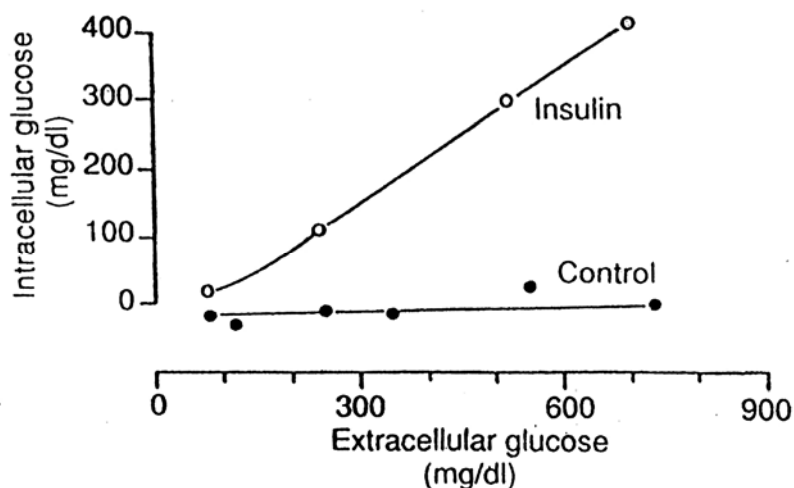


Figure 7.1

- (a) With reference to Figure 7.1, what can be concluded regarding the effects of insulin on muscle cells? [4]

(b) State the homeostatic significance of such an effect. [1]

(c) Blood glucose levels are under the control of the hormones insulin and glucagon. Suggest why blood glucose levels have to be under such strict control. [3]

Total: [8]

Question 8

Sea slugs are marine molluscs. Some are pictured in Figures 8.1, 8.2 and 8.3. Sea slugs live in shallow coastal waters where they graze on animals that form colonies on rocks, like sponges and corals. Sea slugs are preyed upon by fish.

**Figure 8.1****Figure 8.2****Figure 8.3**

Many sea slugs collect poisonous chemicals or defences such as stinging cells from their prey. They then make use of these in their own bodies. Sea slugs containing poisonous chemicals or other defences are generally brightly-coloured.

Sea slugs evolved from sea snails by the loss of the shell. It is thought that being brightly-coloured is an advantage to poisonous animals as predators learn to avoid them.

- (a) Suggest why being poisonous or brightly-coloured is more common in species of sea slug than in sea snails. [2]

The waters of South-Eastern Australia are home to a group of sea slugs found nowhere else.

Ten distasteful species of the genus *Chromodoris* all share a pattern of bright red spots on a white background. This is an example of mimicry.

- (b) Describe how a South-Eastern Australian sea slug population that was originally white might have developed the red-spotted pattern over a period of time. [3]

Total: [5]

Section C: Free-response Questions (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, wherever appropriate.

Your answer must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9** **(a)** Discuss why the eukaryotic genome is considered to be more complex than those of the prokaryotic genome. [8]
- (b)** Compare and contrast the control of gene expression in prokaryotic and eukaryotic cells. [6]
- (c)** Explain how changes at the genome level can lead to cancer. [6]
- 10** **(a)** Compare and contrast the transmission of a nervous signal within a neurone and its transmission between neurones. [10]
- (b)** Predict and explain the effect of a drug that acts as an inhibitor of acetylcholinesterase at a neuromuscular junction. [4]
- (c)** Describe how the endocrine system differs from the nervous system. [6]

END OF PAPER

Name		Index No.		Civics Group	
------	--	-----------	--	--------------	--



DUNMAN HIGH SCHOOL
SENIOR HIGH
2009 PRELIMINARY EXAMINATION
General Certificate of Education Advanced Level
YEAR SIX H2 BIOLOGY (9747)

BIOLOGY

9747 / 03

PAPER 3 Applications

29 September 09 (Tuesday)

1 hour 30 minutes

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST.

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Do not open this booklet until you are told to do so.

Section A:

Consists of 3 Structured Questions

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section B:

Consists of 1 Essay Question

Answer **all** questions.

Write your answers on the separate **writing papers** provided.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE		
Section D	1	/20
	2	/10
	3	/15
Section E	4	/20
Total		/65

This question paper consists of **11** printed pages and **1** blank page (including this cover page).

Section D: Structured Questions (45 marks)

Answer **all** questions in this section.

Question 1

Figure 1.1 shows the main stages in the polymerase chain reaction (PCR).

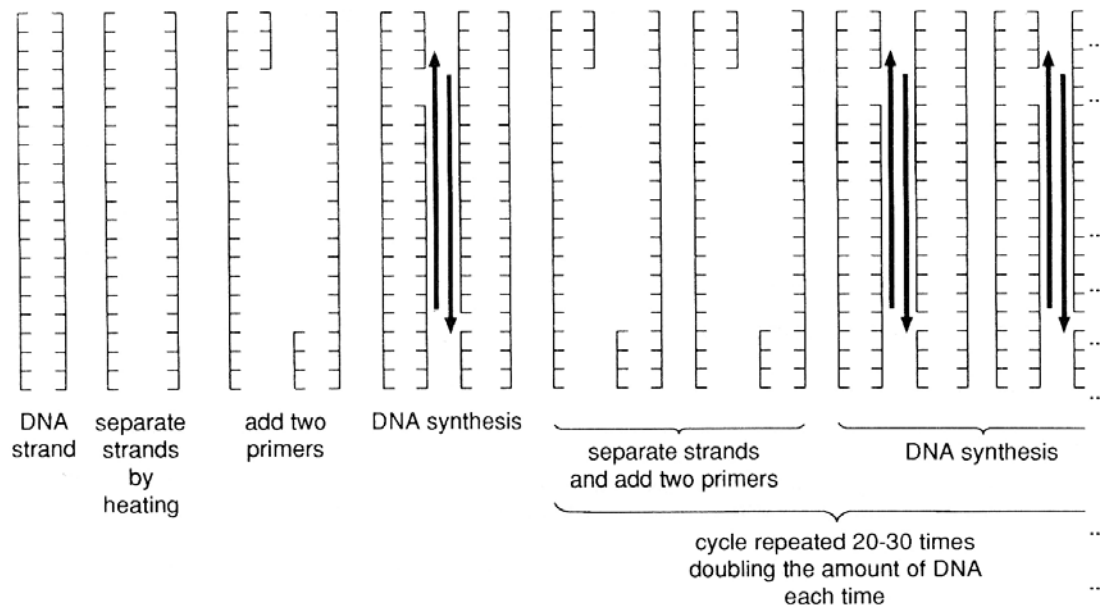


Figure 1.1

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (b)** The products of PCR are often subjected to gel electrophoresis after its reaction.

Discuss the principle of gel electrophoresis. [5]

- (c)** Discuss the differences between cDNA and genomic DNA. [5]

Total: [20]

Question 2

The CFTR gene is found on chromosome 7 and in non-sufferers it gives rise to a protein called Cystic Fibrosis Transmembrane Conductance Regulator (CFTR). The defect which most commonly leads to the disease involves the deletion of three nucleotides from the gene. This is illustrated in Figure 2.1.

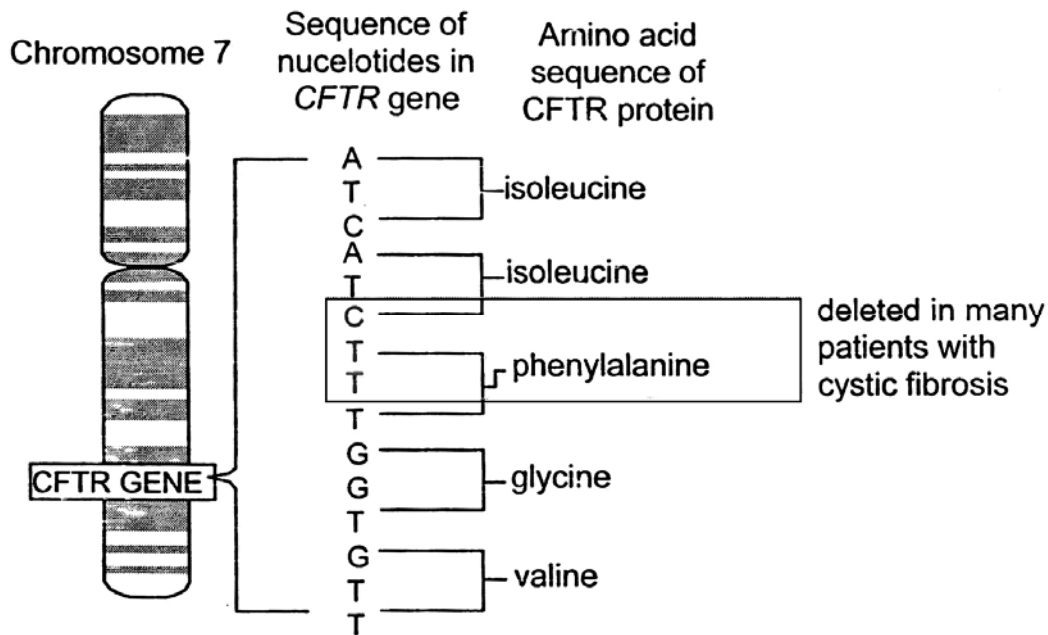


Figure 2.1

- (a) With reference to Figure 2.1, explain how this deletion results in the production of a defective protein. [4]

The protein produced by the CFTR gene is involved in the transport of chloride ions across cell membranes. In sufferers of Cystic Fibrosis (CF) there is a build-up of sodium and chloride ions that causes the body to produce a thick, sticky mucus. Most individuals with the disease have a considerably reduced life expectancy.

- (b) (i)** Outline the problems caused by the defective CFTR gene in sufferers of CF. [3]

- (ii)** Explain why CF usually results in a considerably reduced life expectancy. [3]

Total: [10]

Question 3

- (a) Compare and contrast genetic engineering and traditional selective breeding. [3]

The introduction of genetically engineered or recombinant DNA into pollen grains is made difficult by the pollen's tough outer coat. The walls of the pollen tube that grow out of the germinated grains are only a few molecules thick and are much more easily penetrated.

Pollen grains from the anther of the *Nicotiana glauca* have been successfully genetically engineered. The pollen is put into a culture solution that causes the grains to germinate. Once pollen tubes are produced, recombinant DNA is added and the mixture placed in a tube containing two parallel electrodes, 2 mm apart.

Applying a large electric field between the electrodes in 80 μ s pulses produces tiny holes in the sides of the tubes. The holes remain open for as long as 30 minutes and DNA can enter the pollen cells through the pores. The altered pollen grains are placed on the stigmas of *Nicotiana glauca* plants. Fertilisation takes place, and the resulting seeds are grown and the plants inspected for the production of the introduced gene.

To show that the technique works successfully, a gene from *E. coli* coding for a protein called β -glucuronidase (GUS) has been introduced into *Nicotiana* in this way and GUS has been found in the adult plants.

This technique may be particularly useful for genetically engineering crop plants since genetically modified plants are harder to produce than genetically modified animals.

- (b) Name the technique used in the example above to introduce DNA into the pollen cells [1].

- (c) Give **two** reasons why not all plants produced by this technique will show the product of the introduced gene. [2]

- (d) Suggest **two** reasons why genetically engineered plants are more difficult to produce than genetically engineered animals. [2]

Another method to introduce recombinant DNA into plant cells is via the *Agrobacterium* / host cell system.

Figure 3.1 shows a diagram of the *Agrobacterium* / host cell system. It may well contain details that you have not seen before. It is not to scale.

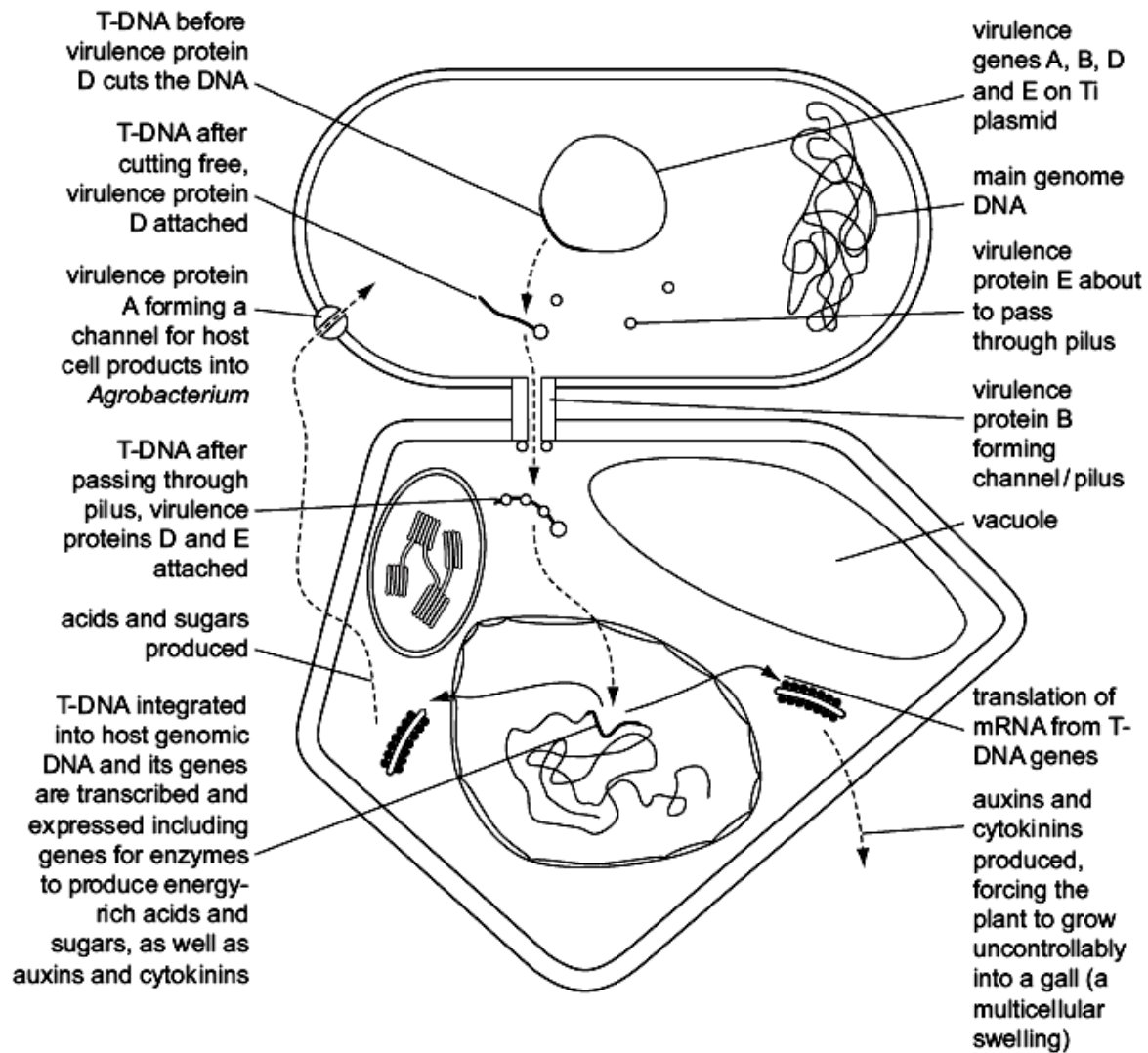


Figure 3.1

- (e) Using information from Figure 3.1 and your knowledge and understanding of genetic engineering, explain how this *Agrobacterium* / host cell system can be used to transform plant cells and produce plants carrying the recombinant DNA. [6]

- (f) Suggest why cloning a plant is easier than cloning an animal. [1]

Total: [15]

Section E: Free-response Questions (20 marks)

Answer **all** questions.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, wherever appropriate.

Your answer must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 **(a)** Discuss the advantages and evolutionary consequences of using plant tissue culture to propagate transgenic plants. [10]
- (b)** Explain, with reference to Bt corn, the significance of genetic engineering in improving the world food situation. [6]
- (c)** Discuss the ethical issues involved with the production of genetically modified organisms. [4]

Total: [20]